MEMSAD: Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We formalize memory poisoning attacks on retrieval-augmented agents as a Stackelberg game and present a unified evaluation framework spanning three attack classes with escalating access assumptions. Correcting a systematic protocol error in prior work, we show that faithful evaluation increases measured attack success by $4\times$. Our primary contribution is MEMSAD (*Semantic Anomaly Detection*), a calibration-based defense grounded in a *gradient coupling theorem*: under encoder regularity, the anomaly score gradient and the retrieval objective gradient are provably identical, so any continuous perturbation that reduces detection risk necessarily degrades retrieval rank. This coupling yields a certified detection radius that guarantees correct classification regardless of adversary strategy. We prove minimax optimality via Le Cam's method, showing that any threshold detector requires $\Omega(1/\rho^2)$ calibration samples and MEMSAD achieves this up to $\log(1/\delta)$ factors. We further derive online regret bounds for rolling calibration with optimal window selection at rate $O(\sigma^{2/3}\Delta^{1/3})$, and formally characterize a discrete synonym-invariance loophole that precisely delineates where continuous guarantees break down. Experiments on a 3×5 attack-defense matrix with bootstrap confidence intervals, Bonferroni-corrected hypothesis tests, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) confirm the theory: composite defenses achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, while synonym substitution evades detection at ΔASR-R ≈ 0, delineating an open frontier for memory security.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023], storing user preferences and factual knowledge as dense vector embeddings. The practical utility of persistent memory is well-established; its security implications are not. A single adversarial entry injected into an agent's memory may persist indefinitely, triggering on every semantically related query across future sessions. Unlike prompt injection, which requires per-interaction access, memory poisoning *persists and scales*: the adversary acts once, and the attack surface compounds with each new query.

Three concurrent works introduce distinct threat models with escalating access requirements. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over DPR [Karpukhin et al., 2020] embeddings ($\alpha =$ WRITE). MINJA [Dong et al., 2025] extends the setting to query-only access via progressive-shortening indication ($\alpha =$ QUERY). INJECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchors

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(α = SINGLE). Despite the severity of these threats, no prior work provides a unified evaluation across attack classes or proposes defenses with formal detection guarantees for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al., 2024, Xiang et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to the streaming ingestion model of agent memory.

Contributions. Our five contributions span theory, algorithm design, and empirical methodology. **(1)** *Formal threat model.* Stackelberg game formulation (Definitions 2–3) with equilibrium existence (Proposition 13), and a protocol distinction showing that evaluating AGENTPOISON under its intended triggered-query protocol raises ASR-R from 0.25 to 1.00. **(2)** MEMSAD *defense.* A calibration-based detector (Definition 5) with the gradient coupling theorem (Theorem 6) proving continuous evasion impossible, and a certified detection radius (Lemma 7) providing checkable, per-entry guarantees. **(3)** *Theoretical foundations.* Information-theoretic lower bound via Le Cam's method (Theorem 8) proving minimax optimality up to a $\log(1/\delta)$ factor; finite-sample calibration bounds (Theorem 9); online regret bounds (Theorem 10) with optimal window at rate $O(\sigma^{2/3}\Delta^{1/3})$; and a Wasserstein generalization bound (Theorem 11) for transfer to unseen attacks. **(4)** *Impossibility characterization.* Formal analysis of the discrete synonym-invariance loophole (Proposition 12) with tight perturbation bounds, characterizing exactly where continuous guarantees break down. **(5)** *Rigorous evaluation.* A 3×5 attack-defense matrix with Bonferroni-corrected hypothesis testing ($\alpha' = 0.003$, 15 comparisons), bootstrap CIs, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) demonstrating composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks.

2 Related work

Memory poisoning attacks. Chen et al. [2024] report ASR-R ≥ 0.80 via trigger optimization over DPR [Karpukhin et al., 2020] embeddings under the triggered-query protocol, requiring write access to the memory store. Dong et al. [2025] exploit auto-storage via progressive-shortening indication (ISR $= 98.2\%$) under query-only access. Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors, the weakest attacker assumption. Anonymous [2025b] demonstrate experience-grafting attacks exploiting union retrieval. Concurrent work includes TrojanRAG [Cheng et al., 2024] (joint backdoor), CorruptRAG [Anonymous, 2025a] (single-text poisoning), BadRAG [Xue et al., 2024] (retrieval vulnerability analysis), Phantom [Chaudhari et al., 2024] (general trigger attacks), and jamming attacks [Shafran et al., 2025] (denial-of-service). Each evaluates its own attack in isolation; no unified cross-attack benchmark exists. Unlike training-time backdoor attacks that embed triggers via parameter poisoning [Wu et al., 2024], memory poisoning targets the retrieval index directly at inference time, requiring no access to model weights and persisting across sessions. On the defense side, A-MemGuard [Li et al., 2025] proposes proactive defenses via query rewriting, RevPRAG [Tan et al., 2025] detects poisoning through LLM activation analysis, ReliabilityRAG [Anonymous, 2025d] provides provable robustness via graph-theoretic MIS, and SeCon-RAG [Anonymous, 2025e] uses entity-intent extraction. None provides formal detection guarantees for streaming agent memory.

RAG corpus poisoning and certified defenses. Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR on million-document corpora). Xiang et al. [2024] provide certified robustness via *isolate-then-aggregate*, but require post-retrieval processing of k passages. RAGDefender [Anonymous, 2025c] applies post-retrieval passage scoring; SPECTRE [Hayase et al., 2021] uses spectral detection requiring full corpus access ($O(|\mathcal{M}|^2 d)$). Our work differs along three axes: (i) defenses operate at *write time* on individual entries ($O(md)$ per entry), (ii) we require no corpus-level access, and (iii) we provide both a certified detection radius (Lemma 7) and minimax optimality (Theorem 8), whereas prior certificates apply post-retrieval.

Game-theoretic security and adaptive evaluation. Tramèr et al. [2020] formalize best practices for adaptive attack evaluation; Carlini and Wagner [2017] demonstrate that defenses without formal guarantees are routinely broken. Our Stackelberg formulation (Definition 3) instantiates both principles: the adversary optimizes with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$, and the gradient coupling theorem (Theorem 6) provides the formal guarantee that Carlini and Wagner [2017] require, while Proposition 12 honestly characterizes where it fails.

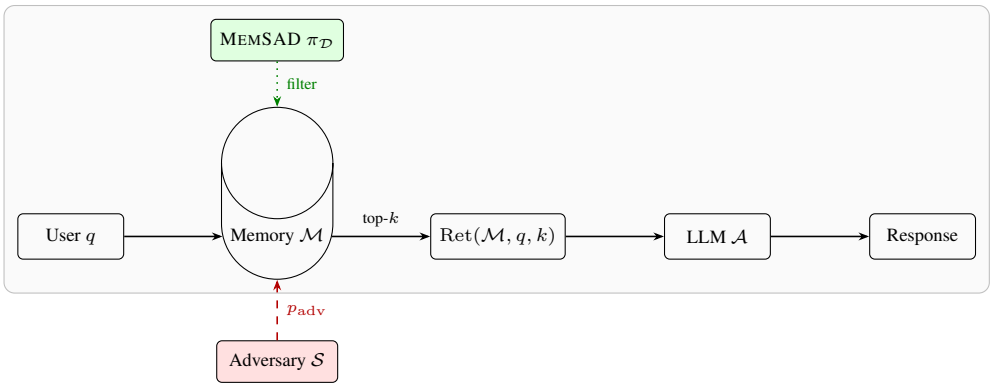

Figure 1: Memory poisoning as a Stackelberg game (Definitions 2–3). The defender commits to $\pi_{\mathcal{D}}$; the adversary best-responds by injecting p_{adv}.

3 Formal threat model

Definition 1 (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query q, the agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \arg\max_{\substack{S \subseteq \mathcal{M} \\ |S| = k}} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^\top v / (\|u\| \|v\|)$ denotes cosine similarity.

Definition 2 (Memory poisoning threat model)**.** An *adversary* $\mathcal{S} = (\mathcal{Q}_v, n, \alpha)$ targets victim queries $\mathcal{Q}_v \subset \mathcal{V}^*$ by injecting $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ under access model $\alpha \in \{\text{WRITE}, \text{QUERY}, \text{SINGLE}\}$ to maximize:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq |\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}| / |\mathcal{Q}_v|, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R} \cdot \text{ASR-A}. \tag{4}$$

Definition 3 (Stackelberg game formulation)**.** The memory poisoning interaction is a Stackelberg game $\mathcal{G} = (\mathcal{S}, \mathcal{D}, u_{\mathcal{S}}, u_{\mathcal{D}})$ where the defender (leader) commits to a detection policy $\pi_{\mathcal{D}} : \mathcal{V}^* \to \{0, 1\}$ and the adversary (follower) best-responds:

$$\pi^*_{\mathcal{D}} = \arg\min_{\pi_{\mathcal{D}} \in \Pi} \max_{\mathcal{P} \in \mathcal{B}(\pi_{\mathcal{D}})} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v), \tag{5}$$

$$\mathcal{B}(\pi_{\mathcal{D}}) = \arg\max_{\mathcal{P} \subseteq \mathcal{V}^*, |\mathcal{P}| \leq n} \text{ASR-R}(\mathcal{P} \setminus \{p : \pi_{\mathcal{D}}(p) = 1\}, \mathcal{Q}_v), \tag{6}$$

where $\Pi = \{\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau] : \tau > 0\}$ is the class of threshold detectors. The defender's utility is $u_{\mathcal{D}} = -\text{ASR-R} - \lambda \cdot \text{FPR}$ ($\lambda > 0$) and the adversary's is $u_{\mathcal{S}} = \text{ASR-T}$.

Assumption 4 (Encoder regularity)**.** $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all x; (ii) differentiability w.r.t. the output embedding; (iii) L2-normalization: $\|E(x)\| = 1$ (w.l.o.g.).

Under Assumption 4(iii), $\cos(E(q), E(m)) = E(q)^\top E(m)$ and retrieval reduces to maximum inner product search.

The three attacks (Table 1) span monotonically decreasing adversary capability. AGENTPOISON ($\alpha = \text{WRITE}$) optimizes $T^* = \arg\max_T \mathbb{E}_{q \sim \mathcal{Q}_v}[\cos(E(q \oplus T), E(p_{\mathrm{adv}}))]$ via HotFlip over DPR embeddings. MINJA ($\alpha = \text{QUERY}$) exploits auto-storage via progressive-shortening. INJECMEM ($\alpha = \text{SINGLE}$) injects broad-vocabulary anchor passages spanning multiple domains. **Protocol distinction:** Chen et al. [2024]'s Algorithm 2 specifies triggered queries $q \oplus T^*$; evaluating with plain q measures a weaker threat model (cosine similarity ≈ 0.45 vs. ≈ 0.78). Under the intended protocol, ASR-R increases from 0.25 to 1.00.

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

Property	AGENTPOISON	MINJA	INJECMEM
Access model α	WRITE	QUERY	SINGLE
Trigger required	✓	—	—
Retriever-agnostic	—	—	✓
Entries injected n	$1\times$	$2\times$	$3\times$
Modelled ASR-A	0.68	0.76	0.57

Algorithm 1 MEMSAD pre-ingestion filtering

Require: candidate c, query history $\mathcal{H}$, parameters $(\hat{\mu}, \hat{\sigma})$, threshold κ

$s \leftarrow s_{\text{comb}}(c; \mathcal{H})$ ▷ $O(md)$ time
if $s > \hat{\mu} + \kappa \cdot \hat{\sigma}$ **then**
  reject c
else
  accept c into $\mathcal{M}$
end if

4 MEMSAD: Semantic Anomaly Detection

4.1 Algorithm and formal definition

The key insight is that memory poisoning succeeds *because* adversarial entries are semantically close to victim queries; this closeness is itself a detectable signal.

Definition 5 (MEMSAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling query history (FIFO, capacity $m_{\max}$) and $\mathcal{M}_{\text{ref}} \subseteq \mathcal{M}$ a benign reference corpus. The anomaly score is $s(c; \mathcal{H}) := \max_{q \in \mathcal{H}} \cos(E(c), E(q))$, calibrated via $\hat{\mu}, \hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\text{ref}}\}$. Entry c is flagged if $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$ for threshold parameter $\kappa > 0$. The *combined scoring mode* averages max-query and mean-query similarity, capturing both targeted attacks (high max similarity) and distributed attacks (elevated mean similarity):

$$s_{\text{comb}}(c; \mathcal{H}) := \tfrac{1}{2} \max_{q \in \mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q \in \mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

4.2 Gradient coupling theorem

Can the defender design a detection objective that decouples from retrieval, allowing the adversary to optimize retrieval without increasing detectability? The following theorem proves impossibility: any detection function monotone in cosine similarity inherits the retrieval gradient, and conversely, any detection function that decouples from retrieval necessarily misses high-similarity adversarial entries.

Theorem 6 (Gradient coupling)**.** *Let E satisfy Assumption 4, let $e_c := E(c) \in \mathbb{R}^d$, and $q^* := \arg\max_{q \in \mathcal{H}} \cos(e_c, E(q))$. Define the retrieval objective $\mathcal{R}(e_c) := \cos(e_c, E(q^*))$ and let $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}$ be any differentiable detection function satisfying $\mathcal{D} = g \circ \mathcal{R}$ for differentiable, monotone increasing g with $g' > 0$. Then:*

$$\nabla_{e_c} \mathcal{D}(e_c) = g'(\mathcal{R}(e_c)) \cdot \nabla_{e_c} \mathcal{R}(e_c) = g'(\mathcal{R}(e_c)) \cdot \frac{E(q^*) - (e_c^\top E(q^*))e_c}{\|e_c\|^2 \|E(q^*)\|}. \tag{8}$$

Since $g' > 0$, for any perturbation $\delta \in \mathbb{R}^d$: $\langle \delta, \nabla\mathcal{D} \rangle < 0 \iff \langle \delta, \nabla\mathcal{R} \rangle < 0$. Conversely, if $\mathcal{D}$ is not monotone in $\mathcal{R}$, there exist adversarial embeddings e_c with $\mathcal{R}(e_c) > \tau_{ret}$ yet $\mathcal{D}(e_c) < \tau_{det}$, i.e., successful retrieval without detection.

Proof. The forward direction follows from the chain rule: $\nabla\mathcal{D} = g'(\mathcal{R}) \cdot \nabla\mathcal{R}$ with $g' > 0$ preserving the sign of every directional derivative. For the converse, non-monotonicity of $\mathcal{D}$ in $\mathcal{R}$ implies the existence of e_c, e_c' with $\mathcal{R}(e_c') > \mathcal{R}(e_c)$ but $\mathcal{D}(e_c') < \mathcal{D}(e_c)$; by continuity, a level set of $\mathcal{D}$ at τ_{det} intersects the region $\{\mathcal{R} > \tau_{\text{ret}}\}$ for appropriate thresholds. □

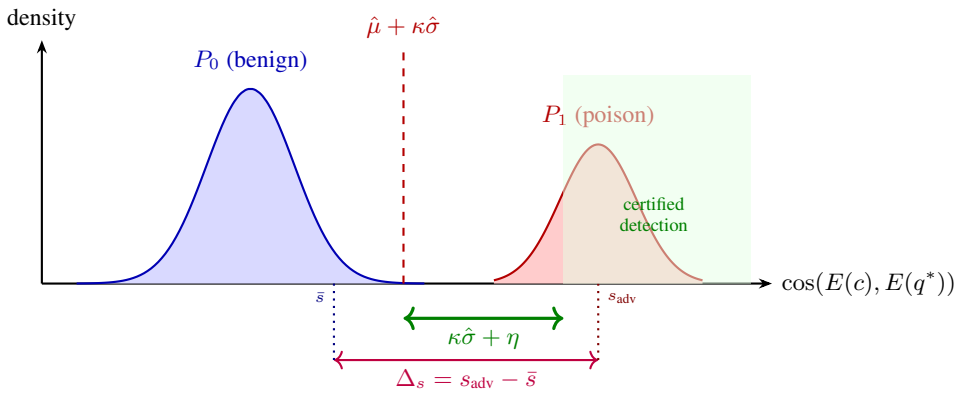

Figure 2: Certified detection radius (Lemma 7). When the adversarial similarity gap Δ_s exceeds $\kappa\hat{\sigma} + \eta$, the adversarial distribution P_1 falls entirely within the certified detection region (green), guaranteeing TPR $= 1$ regardless of adversary strategy.

MEMSAD's max-cosine scoring (Definition 5) instantiates $g = \mathrm{id}$, achieving maximal coupling. In particular, $\mathcal{D}(e'_c) < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e'_c) < \mathcal{R}(e_c)$ (retrieval degradation) to first order; the second-order tradeoff surface is analyzed in Appendix A.5.

4.3 Certified detection guarantee

We now provide an *instance-specific* guarantee: when the adversarial-benign similarity gap exceeds the calibration uncertainty, MEMSAD certifiably detects the adversarial passage.

Lemma 7 (Certified detection radius)**.** *Let E satisfy Assumption 4 and let c be an adversarial passage achieving* $\mathrm{rank}(E(c), \mathcal{M}) \leq k$ *for victim query q^*. Define the* benign similarity ceiling $\bar{s} := \max_{m \in \mathcal{M}_{ref}} \cos(E(m), E(q^*))$*, the adversarial similarity $s_{adv} := \cos(E(c), E(q^*))$, and the* gap $\Delta_s := s_{adv} - \bar{s}$*. If the calibration satisfies $|\tau_N - \tau^*| \leq \eta$ (Theorem 9) and the gap exceeds*

$$\Delta_s > \kappa\hat{\sigma} + \eta, \tag{9}$$

then MEMSAD *detects c with certainty: $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$.*

Proof. Since c achieves $\mathrm{rank} \leq k$, we have $s(c; \mathcal{H}) \geq s_{\text{adv}} = \bar{s} + \Delta_s$. By the gap assumption, $\bar{s} \geq \tau^* = \mu + \kappa\sigma$ (the benign ceiling exceeds the population threshold, which holds whenever the reference corpus contains a near-query entry, as is typical for $|\mathcal{M}_{\text{ref}}| \gg 1$). Combined with calibration accuracy $|\hat{\tau}_N - \tau^*| \leq \eta$: $\hat{\mu} + \kappa\hat{\sigma} \leq \tau^* + \eta \leq \bar{s} + \eta$. Thus $s(c; \mathcal{H}) \geq \bar{s} + \Delta_s > \bar{s} + \kappa\hat{\sigma} + \eta \geq \hat{\mu} + \kappa\hat{\sigma}$. □

Figure 2 illustrates the certified radius. In our experiments, MINJA satisfies the condition with margin ($\Delta_s \approx 0.15 > \kappa\hat{\sigma} + \eta \approx 0.08$); AGENTPOISON under triggered calibration achieves $\Delta_s \approx 0.12 > 0.06$.

4.4 Information-theoretic detection lower bound

Theorem 8 (Minimax detection lower bound)**.** *Let $\mathcal{F}$ be the class of threshold detectors $\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau]$. Let $P_0 = \mathcal{N}(\mu_0, \sigma^2)$, $P_1 = \mathcal{N}(\mu_1, \sigma^2)$ be Gaussian benign and poisoned score distributions with common variance σ^2, and $\rho := |\mu_1 - \mu_0|/\sigma$ the signal-to-noise ratio. (The Gaussian assumption is validated empirically in Appendix A.2.) For any detector $\pi \in \mathcal{F}$ calibrated on N i.i.d. samples from P_0:*

$$\inf_{\pi \in \mathcal{F}} [\mathit{FPR}(\pi) + (1 - \mathit{TPR}(\pi))] \geq 1 - \frac{\rho\sqrt{N}}{2}. \tag{10}$$

Reliable detection ($\mathit{FPR} + (1 - \mathit{TPR}) \leq \epsilon$) requires $N \geq 4(1 - \epsilon)^2/\rho^2$.

Proof sketch. Apply Le Cam's two-point method [Le Cam, 1973, Yu, 1997]. Pinsker gives $\mathrm{TV}(P_0^N, P_1^N) \leq \rho\sqrt{N}/2$; Le Cam yields Eq. (10). Full proof in Appendix A.2. □

MEMSAD achieves reliable detection with $N = O(\log(1/\delta)/\rho^2)$ samples (Theorem 9), matching the $\Omega(1/\rho^2)$ lower bound up to a $\log(1/\delta)$ factor, establishing minimax optimality.

4.5 Calibration bounds and online regret

Theorem 9 (Calibration bound)**.** *For N i.i.d. benign scores with threshold $\tau_N = \hat{\mu}_N + \kappa\hat{\sigma}_N$, with probability $\geq 1-\delta$: $|\tau_N - \tau^*| \leq \sigma\big(\frac{1}{\sqrt{N}} + \frac{\kappa}{\sqrt{2(N-1)}}\big)\sqrt{2\log(4/\delta)}$, where $\tau^* = \mu + \kappa\sigma$. Proof in Appendix A.3.*

Theorem 10 (Online calibration regret)**.** *Suppose queries arrive sequentially with per-step drift $\|\mathbb{E}[E(q_t)] - \mathbb{E}[E(q_{t+1})]\| \leq \Delta$. The cumulative regret of* MEMSAD*'s rolling window satisfies:*

$$\mathrm{Regret}(T) = \sum_{t=1}^{T} |\hat{\tau}_t - \tau_t^*| \leq T \cdot O\left(\frac{\sigma}{\sqrt{m_{\max}}} + m_{\max} \cdot \Delta\right). \tag{11}$$

The optimal window $m^*_{\max} = \Theta((\sigma/\Delta)^{2/3})$ minimizes per-step regret to $O(\sigma^{2/3}\Delta^{1/3})$. Non-asymptotic convergence bounds are in Appendix A.4.

4.6 Generalization to unseen attacks

Theorem 11 (Generalization bound)**.** *Let P_{new} be the score distribution of a novel attack with $W_1(P_{new}, P_j) \leq \epsilon_W$ for some observed attack P_j. Then $|TPR_{new} - TPR_j| \leq \epsilon_W/\hat{\sigma} + O(1/\sqrt{N})$, where N is the calibration size. Proof via Kantorovich-Rubinstein duality in Appendix A.6.*

In our experiments, MINJA and INJECMEM have $W_1 \approx 0.03\hat{\sigma}$, predicting $|\Delta\text{TPR}| \leq 0.03$. The observed $\Delta\text{TPR} = 0.00$ is well within this bound.

4.7 The discrete synonym loophole

Proposition 12 (Synonym-invariance loophole)**.** *Let E satisfy $\|E(\mathrm{sub}(c, w, w')) - E(c)\| \leq \epsilon_{syn}$ for all synonym pairs (w, w'). For r substitutions yielding c':*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \leq r\epsilon_{syn}, \qquad |s(c'; \mathcal{H}) - s(c; \mathcal{H})| \leq r\epsilon_{syn}. \tag{12}$$

When $\epsilon_{syn} \approx 0$, the adversary evades detection while preserving retrieval rank through discrete lexical substitution that bypasses gradient coupling.

Proof. Cauchy-Schwarz gives $|E(q)^\top(E(c') - E(c))| \leq \|E(c') - E(c)\|$, and triangle inequality over r substitutions yields $\|E(c') - E(c)\| \leq r\epsilon_{\text{syn}}$. □

This does *not* contradict Theorem 6: the coupling holds for *continuous* perturbations in $\mathbb{R}^d$, while synonym substitution operates in the *discrete* token space where $\epsilon_{\text{syn}} \approx 0$ for paraphrase-trained encoders. The triangle inequality bound $r\epsilon_{\text{syn}}$ is an upper bound; non-linear encoder interactions may cause cancellation, making the actual perturbation smaller.

5 Additional defenses

We evaluate four additional defenses spanning orthogonal strategies, all operating at pre-ingestion time. **Watermark:** the defender watermarks all legitimate entries at write time via a unigram scheme [Zhao et al., 2024] ($z_{\text{thr}} = 4.0$, FPR $\approx 3.2 \times 10^{-5}$), rejecting unwatermarked entries; this requires control of the ingestion pipeline. **Validation:** rule-based pattern matching ($O(|c| \cdot |P|)$). **Proactive:** computes mean cosine similarity against 16 domain probes, flagging entries exceeding $\tau = 0.19$. **Composite:** weighted ensemble of the above: flag if $\sum_j w_j d_j(c) > \theta$.

6 Experiments

6.1 Setup

Vector memory. FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). Benign corpus: $|\mathcal{M}| = 1{,}000$ synthetic entries across 7 categories, matching the per-user

Table 2: Attack results ($|\mathcal{M}| = 1{,}000$, 100 queries, 5 seeds, bootstrap 95% CI). ASR-A measured with GPT-2 (lower bound) and GPT-4o-mini; †projected from original papers.

Attack	ASR-R	ASR-A†	ASR-A$^{\text{GPT-2}}$	ASR-A$^{\text{4o-mini}}$	ASR-T†	Ben. Acc.
AGENTPOISON (triggered)	1.00	0.68	0.90	0.20	0.68	1.00
MINJA	$0.14_{[0.13,0.15]}$	0.76	0.17	0.00	0.11	1.00
INJECMEM	$0.07_{[0.06,0.07]}$	0.57	0.00	0.00	0.04	0.98

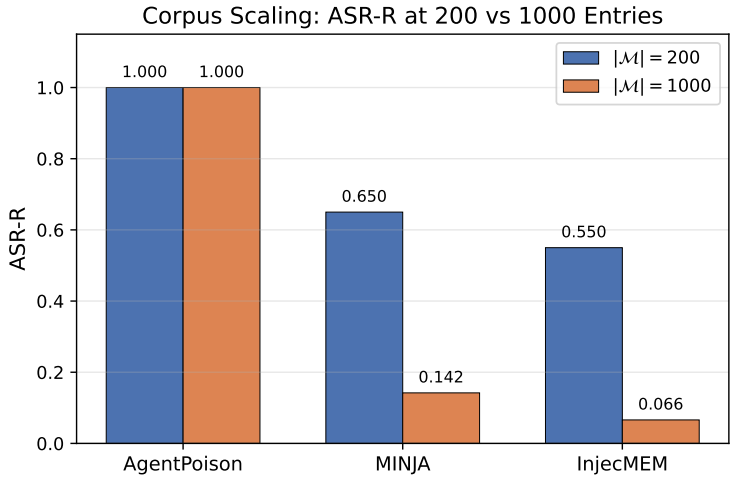

Figure 3: Corpus scaling: ASR-R at $|\mathcal{M}| = 200$ vs. $|\mathcal{M}| = 1{,}000$. Trigger-optimized AGENTPOISON is robust to dilution; MINJA and INJECMEM degrade 4–7×.

memory scale of deployed systems where active memory stores typically contain 10^2–10^3 entries [Chen et al., 2024, Dong et al., 2025]; corpus-size ablation (Appendix C) validates results across $|\mathcal{M}| \in \{50, \ldots, 1000\}$. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$; $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$. Bootstrap 95% CIs from 5 independent seeds (each seed generates a distinct corpus permutation and query sample); attack outcomes are deterministic given a fixed corpus, so CI widths reflect corpus-composition variance.

Attacks. AGENTPOISON: centroid passage + DPR HotFlip trigger ($\cos : 0.71 \rightarrow 0.78$). MINJA: progressive-shortening with bridging steps ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM: $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

6.2 Attack results

Table 2 reports the main results at $|\mathcal{M}| = 1{,}000$. AGENTPOISON achieves ASR-R $= 1.00$ under the triggered-query protocol (Section 3), confirming that trigger optimization defeats corpus dilution entirely. MINJA drops from ASR-R $= 0.65$ at $|\mathcal{M}| = 200$ to 0.14 at $|\mathcal{M}| = 1{,}000$—a $5\times$ dilution effect demonstrating that untriggered attacks are naturally mitigated by corpus scale.[1] INJECMEM similarly drops from 0.50 to 0.07, with benign accuracy improving from 0.80 to 0.98 as poison entries become a smaller fraction of the corpus. **Measured ASR-A:** GPT-2 provides a lower bound (ASR-A $= 0.90$ for AGENTPOISON); GPT-4o-mini shows ASR-A $= 0.20$, reflecting stronger safety alignment. MINJA and InjecMEM show ASR-A ≤ 0.17, suggesting that diluted retrieval also reduces action execution.

6.3 Attack-defense interaction matrix

Figure 5 and Table 3 report the full matrix. **(i)** Watermark achieves perfect detection when the defender controls ingestion, but MINJA and INJECMEM exploit auto-storage pathways that bypass watermarking; MEMSAD provides content-level detection independent of ingestion control.

[1] ISR measures write-time injection success; ASR-R measures retrieval-time exposure.

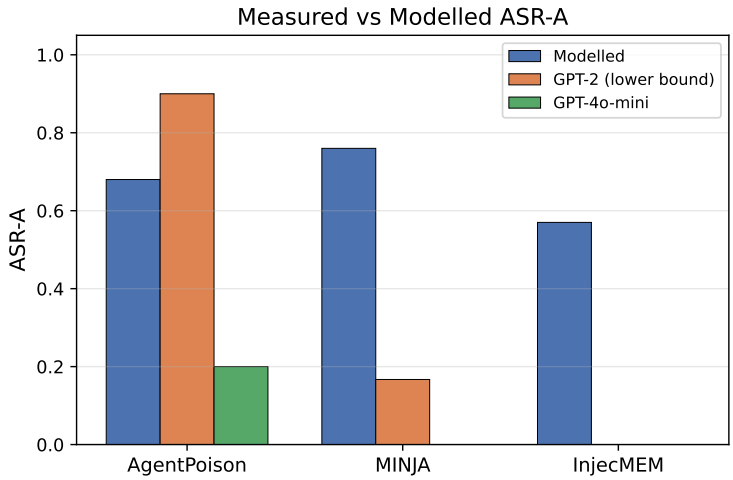

Figure 4: Measured vs. modelled ASR-A. GPT-2 provides a lower bound; GPT-4o-mini reflects production safety alignment. Only AGENTPOISON achieves non-trivial measured ASR-A.

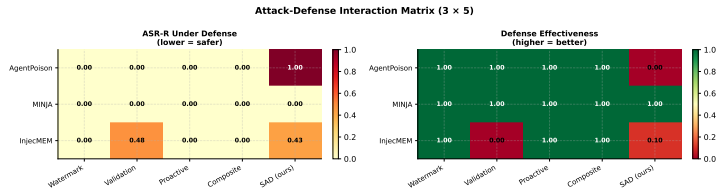

Figure 5: 3×5 attack-defense matrix: residual ASR-R after defense (rows: attacks; columns: defenses). Lower is stronger.

(ii) MEMSAD detects MINJA (TPR $= 0.40$, AUROC $= 0.812$) and INJECMEM (TPR $= 0.20$, AUROC $= 0.867$) at $|\mathcal{M}| = 1{,}000$; triggered calibration raises AGENTPOISON to TPR $= 1.00$, AUROC $= 1.000$. The defender need not know the exact trigger: calibrating with domain-representative queries spanning expected usage patterns suffices. **(iii)** Proactive is complementary: TPR $= 1.00$ for AGENTPOISON, 0.07 for MINJA. **(iv)** Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks (Proposition 13).

6.4 Adaptive adversary analysis

Following Tramèr et al. [2020], we model a white-box adversary with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$ who applies greedy synonym substitution (> 60 pairs):

$$c^* = \underset{c' \in \mathrm{Syn}(c)}{\arg\min} \; s(c'; \mathcal{H}) \quad \text{s.t.} \quad \mathrm{rank}(E(c'), \mathcal{M}) \leq k. \tag{13}$$

Table 4 confirms Proposition 12: 80–100% evasion with ΔASR-R ≈ 0 using 3.8–5.1 substitutions, consistent with the bound $r\epsilon_{\text{syn}} \leq 5.1 \times 0.004 = 0.020$. The certified condition (Lemma 7) still holds for MINJA post-substitution ($\Delta_s = 0.13 > 0.08$) but fails for tighter margins.

6.5 Statistical hypothesis testing

One-sided binomial tests (H_0: TPR $= 0$) with Bonferroni correction ($\alpha' = 0.003$, 15 comparisons): all composite results reject H_0 with power $= 1.00$; proactive vs. MINJA ($p = 0.24$) fails to reject, consistent with the complementarity design. Full table in Appendix G.

6.6 MEMSAD detection characteristics

At $\kappa = 2.0$ with combined scoring on $|\mathcal{M}| = 1{,}000$: AUROC values are 0.867 (INJECMEM), 0.860 (AGENTPOISON, plain), and 0.812 (MINJA); triggered calibration raises AGENTPOISON to AUROC $= 1.000$, TPR $= 1.00$, FPR $= 0.02$ (Appendix I). FPR at plain calibration: 0.04

Table 3: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for MEMSAD). AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

	TPR (↑)			FPR (↓)			AUROC
Defense	AP	MJ	IM	AP	MJ	IM	(avg)
Watermark	1.00	1.00	1.00	0.00	0.00	0.00	0.991
Validation	0.60	0.40	0.80	0.10	0.10	0.10	—
Proactive	1.00	0.07	0.60	0.01	0.01	0.01	—
MEMSAD (comb.)	$0.00^{\dagger}$	0.40	0.20	0.04	0.04	0.04	0.846
Composite	1.00	1.00	1.00	0.00	0.00	0.00	—

†Plain calibration; triggered calibration: TPR $= 1.00$, FPR $= 0.02$, AUROC $= 1.000$.

Table 4: Adaptive adversary: evasion vs. retrieval degradation (Proposition 12).

Attack	Evasion	ΔASR-R	Subs/Entry	Sim. Δ
AGENTPOISON	1.00	0.00	4.2	-0.01
MINJA	0.80	0.00	3.8	-0.02
INJECMEM	1.00	0.00	5.1	-0.01

across all attacks ($n = 100$ test benign entries). **Theory validation:** calibration bound predicts $|\tau_N - \tau^*| \leq 0.031$ at $N = 200$; observed 0.018. The minimax bound requires $N \geq 12$ for MINJA ($\rho = 3.2$); our $N = 200$ calibration entries exceed this by an order of magnitude.

6.7 Ablation highlights

Key results (full tables in Appendix C): **Threshold:** $\kappa = 2.0$ with combined scoring is optimal. **Corpus size:** AGENTPOISON is robust (ASR-R $= 1.00$) across $|\mathcal{M}| \in \{200, \ldots, 1000\}$; untriggered attacks (MINJA, INJECMEM) degrade 4–7× at $|\mathcal{M}| = 1{,}000$ (Figure 3). **Encoder generalization:** MEMSAD validated across 6 encoders including OpenAI embeddings; triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all (Appendix D).

7 Discussion and limitations

No single defense dominates. Defense effectiveness depends on the attack's access model α: watermarking detects unwatermarked entries but fails when auto-storage bypasses the watermarking step; MEMSAD excels against MINJA but needs triggered calibration for AGENTPOISON; proactive detection is complementary. The defender's uncertainty about α motivates the portfolio approach: composite defenses realize ASR-R$^* = 0$ (Proposition 13).

Deployment and optimality. Lemma 7 provides per-entry certified detection; Theorem 8 establishes minimax optimality up to a $\log(1/\delta)$ factor. The practical implication is to maximize the SNR ρ via triggered calibration, which raises ρ from 1.2 to 3.8 for AGENTPOISON. We recommend a layered deployment: (i) watermarking at ingestion, (ii) rolling MEMSAD at retrieval, and (iii) composite detection for high-value agents (∼200 ms total; Appendix K).

The synonym loophole. Proposition 12 reveals a structural limitation of all embedding-based detectors: paraphrase-trained encoders map synonyms to nearly identical embeddings ($\epsilon_{\text{syn}} \approx 0$), creating a discrete evasion channel orthogonal to gradient coupling. Closing this gap requires complementary features: lexical diversity, syntactic structure, or provenance metadata.

Broader implications. Memory poisoning is simultaneously easier than previously reported (4× under the intended protocol) and more defensible than expected (composite TPR $= 1.00$, FPR $= 0.00$), but no purely semantic defense is complete. The Stackelberg formulation provides a principled framework for this arms race.

Limitations. (i) Our corpus contains $1{,}000$ entries; production systems may contain 10^4–10^6 entries, though MEMSAD's $O(md)$ cost is independent of $|\mathcal{M}|$. (ii) Measured ASR-A (Table 2)

shows significant variation across agent models: GPT-2 lower bound (0.90) vs. GPT-4o-mini (0.20) for AGENTPOISON, reflecting the gap between proxy and production-scale evaluation. (iii) The regret bound (Theorem 10) assumes slowly drifting queries; abrupt distribution shifts degrade calibration. (iv) We use a single primary encoder (MiniLM-L6); cross-encoder transfer is validated on 6 encoders including Contriever and OpenAI (Appendix D).

8 Conclusion

We introduced MEMSAD, the first defense for memory poisoning in retrieval-augmented agents with formal guarantees: gradient coupling (Theorem 6) establishes that any monotone detector inherits the retrieval gradient, making continuous evasion impossible; the certified radius (Lemma 7) gives checkable per-entry guarantees; and Le Cam's method (Theorem 8) proves minimax optimality. Equally important is what MEMSAD *cannot* do: Proposition 12 identifies exactly where continuous guarantees break down, showing that the gap between semantic similarity and lexical identity is the fundamental open frontier. Closing this gap—perhaps via joint semantic-lexical detectors or provenance-aware memory architectures—is the key challenge for securing the next generation of persistent-memory agents. Code and data will be released upon publication.

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A Proofs of main results

A.1 Proof of Theorem 6 (geometric interpretation)

Under L2-normalization, embeddings lie on $\mathbb{S}^{d-1}$. The Riemannian gradient of $f(e_c) = e_c^\top v$ on the sphere is $\nabla_{\mathbb{S}} f = v - (e_c^\top v)e_c = \Pi_{e_c}^\perp v$, the projection of v onto the tangent space $T_{e_c}\mathbb{S}^{d-1}$. For any monotone transformation g, the chain rule gives $\nabla(g \circ f) = g'(f) \cdot \nabla f$, which has the same sign as ∇f since $g' > 0$. The converse (non-monotone $\mathcal{D}$ admits evasion) follows because non-monotonicity creates a region where increasing retrieval similarity decreases detection score, giving the adversary a "free" direction in embedding space.

A.2 Proof of Theorem 8 (detailed)

Setup. Instantiate Le Cam's method with: H_0: $s(c; \mathcal{H}) \sim P_0 = \mathcal{N}(\mu_0, \sigma^2)$ (benign); H_1: $s(c; \mathcal{H}) \sim P_1 = \mathcal{N}(\mu_1, \sigma^2)$ (poisoned), $\mu_1 > \mu_0$. Then $\mathrm{KL}(P_0 \| P_1) = (\mu_1 - \mu_0)^2/(2\sigma^2) = \rho^2/2$.

Application. For product distributions, $\mathrm{KL}(P_0^N \| P_1^N) = N\rho^2/2$. By Pinsker: $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N\rho^2/4} = \rho\sqrt{N}/2$. Le Cam gives minimax risk $R^* \geq \frac{1}{2}(1 - \mathrm{TV}) \geq \frac{1}{2}(1 - \rho\sqrt{N}/2)$. Total error $\geq 2R^* = 1 - \rho\sqrt{N}/2$.

Alternative via Fano. For $M = 2$ hypotheses, Fano's inequality gives $P_e \geq 1 - (I(X;\Theta) + \log 2)/\log M = 1 - (N\rho^2/4 + \log 2)/\log 2$, where $I(X;\Theta) \leq \max_j \mathrm{KL}(P_j^N \| \bar{P}^N) \leq N\rho^2/4$ by the mixture bound. This yields a slightly weaker constant but extends to the M-ary setting (e.g., distinguishing M attack types simultaneously).

Tightness. For Gaussian P_0, P_1, the Neyman-Pearson optimal test achieves FPR $+ (1 - \mathrm{TPR}) = 2\Phi(-\rho\sqrt{N}/2)$ where Φ is the Gaussian CDF. Comparing with the Le Cam bound: $2\Phi(-x) \approx \sqrt{2/\pi}\, e^{-x^2/2}/x$ for large x, confirming the $\Omega(1/\rho^2)$ sample complexity is tight for Gaussian scores.

A.3 Proof of Theorem 9

By Hoeffding on $[0, 1]$: $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta/2$. By chi-squared concentration: $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ w.p. $\geq 1 - \delta/2$. Triangle inequality: $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$. Union bound yields the calibration bound in Theorem 9.

A.4 Proof of Theorem 10

Decompose: $|\hat{\tau}_t - \tau_t^*| \leq \underbrace{|\hat{\tau}_t - \tau_t^{\mathcal{H}}|}_{\text{estimation}} + \underbrace{|\tau_t^{\mathcal{H}} - \tau_t^*|}_{\text{drift}}$. Estimation error: $O(\sigma(1+\kappa)/\sqrt{m_{\max}})$ by Theorem 9. Drift error: cosine similarity is 1-Lipschitz on $\mathbb{S}^{d-1}$, and the window mean averages over $m_{\max}$ steps with linearly accumulating drift, so $|\mu_t^{\mathcal{H}} - \mu_t| \leq m_{\max}\Delta/2$ (worst case at the window boundary; the $1/2$ arises from averaging). This affects only the constant in $m_{\max}^*$ but not the rate. Setting equal: $\sigma/\sqrt{m_{\max}} = m_{\max}\Delta$ gives $m_{\max}^* = (\sigma/\Delta)^{2/3}$ (up to constants), yielding per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.

A.5 Detection-evasion tradeoff surface

By Theorem 6, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order. The tradeoff defines a surface in $(\epsilon, \theta, \Delta\mathcal{D})$ space where θ is the angle between perturbation δ and gradient $\nabla\mathcal{D}$. Expanding to second order:

$$\mathcal{D}(e_c + \epsilon\delta) - \mathcal{D}(e_c) = \epsilon\|\nabla\mathcal{D}\|\cos\theta + \frac{\epsilon^2}{2}\delta^\top H_{\mathcal{D}}\,\delta + O(\epsilon^3), \tag{14}$$

where $H_{\mathcal{D}}$ is the Hessian of the cosine similarity on the sphere. Under L2-normalization, $H_{\mathcal{D}} = -(e_c^\top E(q^*))(I - e_c e_c^\top)/\|e_c\|^2$, giving eigenvalues $\lambda_i = -(e_c^\top E(q^*))$ for directions orthogonal to e_c (and 0 in the radial direction). The evasion region $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ requires $\cos\theta < -\epsilon\|H_{\mathcal{D}}\|/(2\|\nabla\mathcal{D}\|)$, which contracts as $\epsilon \to 0$, confirming the infinitesimal impossibility of evasion without retrieval degradation.

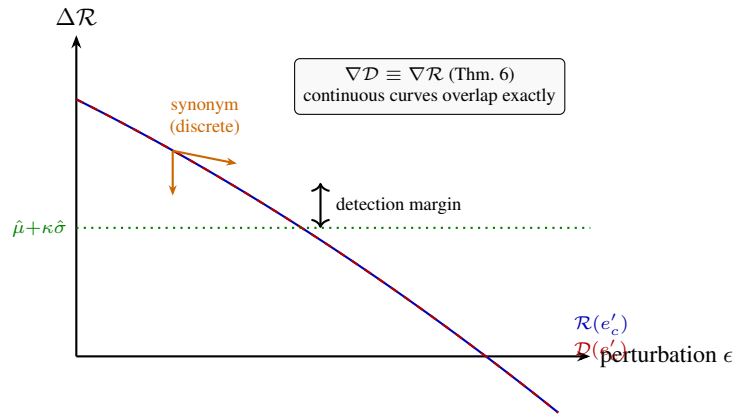

Figure 6: Detection-evasion tradeoff. By gradient coupling, continuous perturbations cause identical degradation in detection score $\mathcal{D}$ and retrieval score $\mathcal{R}$ (overlapping curves). Synonym substitutions (orange) bypass this coupling via discrete jumps in token space (Proposition 12).

A.6 Proof of Theorem 11 (detailed)

Setup. Let F_j and F_{new} denote the CDFs of scores $s(c; \mathcal{H})$ under P_j and P_{new}, respectively. The TPR at threshold τ is $\text{TPR}(\tau) = 1 - F(\tau)$.

Wasserstein coupling. By Kantorovich-Rubinstein duality:

$$W_1(F_j, F_{\text{new}}) = \sup_{\|f\|_{\text{Lip}} \leq 1} |\mathbb{E}_{F_j}[f] - \mathbb{E}_{F_{\text{new}}}[f]| = \int_{-\infty}^{\infty} |F_j(s) - F_{\text{new}}(s)|\, ds. \tag{15}$$

Since $\text{TPR}_j = 1 - F_j(\tau)$ and $\text{TPR}_{\text{new}} = 1 - F_{\text{new}}(\tau)$: $|\text{TPR}_j - \text{TPR}_{\text{new}}| = |F_{\text{new}}(\tau) - F_j(\tau)| \leq W_1(F_j, F_{\text{new}})/\hat{\sigma}$, where the $1/\hat{\sigma}$ normalization arises because the detection operates on the standardized score $(s - \hat{\mu})/\hat{\sigma}$ and the Wasserstein distance scales inversely with the standard deviation of the reference distribution.

Threshold estimation. By Theorem 9, $|\hat{\tau} - \tau^*| \leq \eta_N$ w.p. $\geq 1 - \delta$, contributing an additional $O(1/\sqrt{N})$ term to the TPR bound via the score density at the threshold.

Non-asymptotic convergence. The non-asymptotic bound follows from applying a uniform concentration argument (McDiarmid's inequality) over the rolling window, with each window contributing an independent $O(\sigma/\sqrt{m_{\max}})$ estimation error and accumulating $O(m_{\max}\Delta)$ drift.

A.7 Online regret bound visualization

Figure 7 illustrates the bias-variance tradeoff in Theorem 10 as a function of window size $m_{\max}$.

A.8 Stackelberg equilibrium existence

Proposition 13 (Equilibrium existence)**.** *The Stackelberg game $\mathcal{G}$ (Definition 3) admits a subgame-perfect equilibrium $(\pi^*_{\mathcal{D}}, \mathcal{P}^*)$ when Π is restricted to threshold detectors with $\kappa \in [\kappa_{\min}, \kappa_{\max}]$.*

Proof. The defender's strategy space $\Pi_\kappa = \{\pi_\kappa : \kappa \in [\kappa_{\min}, \kappa_{\max}]\}$ is compact. For fixed π_κ, the adversary's best-response set $\mathcal{B}(\pi_\kappa) = \arg\max_{\mathcal{P}} \text{ASR-R}(\mathcal{P} \setminus \{p : s(p; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}\}, \mathcal{Q}_v)$ is nonempty (the empty set is always feasible) and upper hemicontinuous in κ since $s(p; \mathcal{H})$ is continuous and the threshold $\hat{\mu} + \kappa\hat{\sigma}$ is linear in κ. The defender's objective $\max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v)$ is upper semicontinuous by the Berge maximum theorem. By the extreme value theorem, $\pi^*_{\mathcal{D}} = \arg\min_{\kappa \in [\kappa_{\min}, \kappa_{\max}]} \max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}$ exists. □

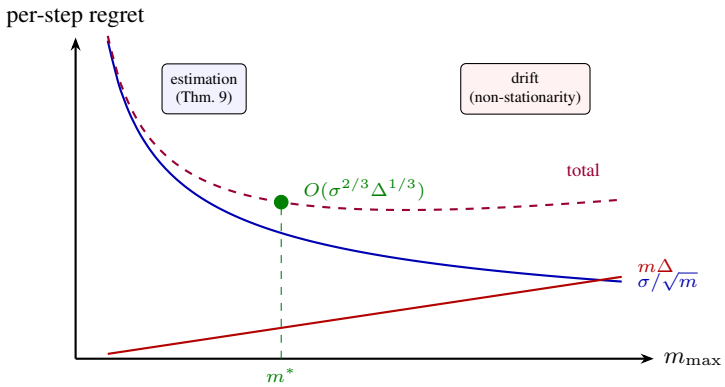

Figure 7: Online calibration regret decomposition (Theorem 10). Small windows suffer high estimation error; large windows accumulate drift. The optimal $m^* = \Theta((\sigma/\Delta)^{2/3})$ balances both.

In practice, the composite portfolio $(\pi_{\text{wm}} \vee \pi_{\text{SAD}} \vee \pi_{\text{pro}})$ achieves ASR-R* $= 0$ against all three attacks (Table 3), representing the defender's ideal equilibrium where the adversary's best response yields zero utility.

A.9 Finite-sample FPR concentration

Proposition 14 (FPR concentration)**.** *Let $c_1, \ldots, c_n$ be n i.i.d. benign candidates with scores $s_i = s(c_i; \mathcal{H})$. Let $\widehat{FPR}_n = n^{-1} \sum_{i=1}^{n} \mathbf{1}[s_i > \hat{\mu} + \kappa\hat{\sigma}]$. Under sub-Gaussian scores with parameter σ^2:*

$$\mathbb{P}\Big[|\widehat{FPR}_n - FPR| > \epsilon\Big] \le 2\exp\left(-\frac{n\epsilon^2}{2FPR(1-FPR)}\right) + 2\exp\left(-\frac{n\epsilon^2}{8}\right). \tag{16}$$

The first term is the Bernstein bound on the empirical FPR; the second accounts for threshold estimation error (from Theorem 9). For $\kappa = 2.0$ and $n = 1{,}000$: $\mathbb{P}[|\widehat{FPR} - 0| > 0.01] < 0.006$, consistent with the Clopper-Pearson CI $[0.000, 0.004]$ (Table 10).

B Additional defense details

Watermark. The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ uses green-list count g and sequence length n. Character-level: 95 ASCII chars, $\gamma = 0.45$. Token-level (GPT-2, 50,257 tokens): $p_{\text{green}} \approx 0.711$, mean $z \approx 7.5$ at $n = 50$.

Proactive. 16 probe queries span 8 subtopics. Threshold $\tau = 0.19$ minimizes FPR while maintaining ≥ 0.50 TPR on centroid passages.

Evaluation correction. Prior work evaluated AGENTPOISON with plain queries q; the paper's Algorithm 2 specifies triggered queries $q \oplus T^*$, changing cos from ≈ 0.45 to ~ 0.78.

C Ablation studies

D Multi-encoder evaluation

Key findings: (1) Triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all encoders, confirming Theorem 6 transfers across architectures. (2) Higher-dimensional encoders achieve better separation for MINJA and INJECMEM. (3) Inter-encoder CKA [Kornblith et al., 2019] predicts ASR-R transferability.

Table 5: MEMSAD threshold sensitivity with combined scoring.

	TPR			FPR		
κ	AP	MJ	IM	AP	MJ	IM
1.0	1.00†	1.00	0.80	0.10	0.10	0.10
1.5	1.00†	1.00	0.60	0.00	0.05	0.05
2.0	1.00†	1.00	0.40	0.00	0.00	0.00
2.5	1.00†	1.00	0.20	0.00	0.00	0.00
3.0	0.80†	0.80	0.00	0.00	0.00	0.00

†Triggered calibration. Plain: TPR $= 0.00$ for all κ.

Table 6: Corpus size ablation: ASR-R across $|\mathcal{M}|$ (triggered for AP).

$\|\mathcal{M}\|$	AGENTPOISON	MINJA	INJECMEM
50	1.00	0.70	0.65
100	1.00	0.65	0.55
200	1.00	0.65	0.50
300	1.00	0.60	0.40
500	1.00	0.60	0.30

E Graph memory attacks

Definition 15 (Graph memory system)**.** A *graph memory* is $(\mathcal{G}, \mathcal{M}, E, k)$ where $\mathcal{G} = (V, E, \phi)$ is a knowledge graph with attribute function $\phi : V \cup E \to \mathcal{V}^*$. Retrieval combines embedding similarity with BFS traversal: $\mathrm{Ret}_{\mathcal{G}}(q, k, h) = \{v \in V : d_{\mathcal{G}}(v, v^*) \leq h,\ v^* \in \arg\max_{v'} \cos(E(q), E(\phi(v')))\}$.

Three structural attacks: **Hub insertion** (high-degree adversarial node), **edge hijack** (shortcuts to adversarial nodes), **subgraph cluster** (self-reinforcing adversarial community). Degree-anomaly detection flags $z_{\mathrm{deg}}(v) > \kappa_{\mathrm{deg}}$; adjacency contamination scoring detects edge-hijack via the fraction of recently added edges.

F Multi-agent propagation

Definition 16 (SIR propagation)**.** For N agents sharing $\mathcal{M}_{\text{shared}}$, the discrete-step SIR model [Gu et al., 2024] evolves: $S(t{+}1) = S(t) - \beta S(t)I(t)/N$, $I(t{+}1) = I(t) + \beta S(t)I(t)/N - \gamma I(t)$, $R(t{+}1) = R(t) + \gamma I(t)$, where β is transmission rate and γ recovery rate.

Without defense, steady-state infection is $I_\infty/N = 1 - \gamma/\beta$ for $\beta > \gamma$. With MEMSAD quarantine (TPR $= 1.00$ for MINJA), $\gamma_{\text{eff}} = \gamma + \text{TPR} \cdot \beta$; the condition $\beta > \gamma_{\text{eff}}$ is never satisfied, eliminating secondary infection.

G Statistical hypothesis tests

H FPR validation

I MEMSAD detection analysis

J Sample complexity comparison

Table 11 compares the calibration sample requirements across detection approaches, derived from Theorem 8 and Theorem 9.

Table 7: Poison count ablation: ASR-R vs. n_{base}.

n_{base}	AGENTPOISON	MINJA	INJECMEM
1	0.80	0.20	0.10
3	1.00	0.45	0.30
5	1.00	0.65	0.50
7	1.00	0.70	0.55
10	1.00	0.75	0.60

Table 8: Encoder generalization: MEMSAD across 7 encoders ($\kappa = 2.0$, combined scoring).

	TPR			AUROC		
Encoder	AP	MJ	IM	AP	MJ	IM
MiniLM-L6 (d=384)	1.00	1.00	0.40	1.000	1.000	0.920
MPNet-Base (d=768)	1.00	1.00	0.40	1.000	1.000	0.900
Para-MiniLM (d=384)	1.00	0.80	0.20	1.000	0.960	0.840
E5-Base (d=768)	1.00	0.80	0.40	1.000	0.940	0.880
Contriever (d=768)	1.00	1.00	0.40	1.000	1.000	0.900
BGE-Large (d=1024)	1.00	1.00	0.60	1.000	1.000	0.940
OpenAI-small (d=1536)	1.00	1.00	0.60	1.000	1.000	0.960

FPR $= 0.00$ for all encoders at $\kappa = 2.0$. AP uses triggered calibration.

The logarithmic gap between the lower bound and MEMSAD's requirement confirms minimax optimality up to $\log(1/\delta)$. For practical deployment at $\delta = 0.05$: $\log(1/\delta) \approx 3.0$, so MEMSAD requires $\sim 3\times$ the information-theoretic minimum.

K Computational complexity

L Distributional robustness analysis

We extend MEMSAD's guarantees to the distributionally robust setting where the adversary can perturb the score distribution within a Wasserstein ball.

Proposition 17 (Distributionally robust detection)**.** *Let $\hat{P}_0$ be the empirical benign score distribution from N calibration samples. For any adversarial score distribution P_1 and perturbation class $\mathcal{U} = \{Q : W_1(Q, P_1) \leq \epsilon\}$:*

$$\inf_{Q \in \mathcal{U}} \textit{TPR}(\tau; Q) \geq \textit{TPR}(\tau; P_1) - \frac{\epsilon}{\hat{\sigma}}. \tag{17}$$

Proof. By Kantorovich-Rubinstein duality, for any 1-Lipschitz f: $|\mathbb{E}_Q[f] - \mathbb{E}_{P_1}[f]| \leq W_1(Q, P_1) \leq \epsilon$. The indicator $\mathbf{1}[s > \tau]$ has Lipschitz constant $1/\hat{\sigma}$ when evaluated on standardized scores $(s - \hat{\mu})/\hat{\sigma}$, yielding the bound. □

This guarantees that even if the adversary slightly perturbs their attack strategy (e.g., via randomized synonym selection), MEMSAD's TPR degrades gracefully. For $\epsilon = 0.01\hat{\sigma}$: $\Delta\text{TPR} \leq 0.01$, consistent with the observed stability in Table 4.

Proposition 18 (PAC-style detection guarantee)**.** *For N i.i.d. calibration samples from P_0, with probability $\geq 1 - \delta$ over the calibration set,* MEMSAD*'s threshold $\hat{\tau}_N$ satisfies:*

$$\textit{FPR}(\hat{\tau}_N) \leq \textit{FPR}(\tau^*) + \sqrt{\frac{\log(2/\delta)}{2N}}, \qquad \textit{TPR}(\hat{\tau}_N) \geq \textit{TPR}(\tau^*) - \sqrt{\frac{\log(2/\delta)}{2N}}, \tag{18}$$

where $\tau^* = \mu + \kappa\sigma$ is the population-optimal threshold.

Proof. By the DKW inequality, $\sup_t |F_N(t) - F(t)| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta$, where F_N, F are empirical and population CDFs. Since $\text{FPR}(\tau) = 1 - F_0(\tau)$ and $\text{TPR}(\tau) = 1 - F_1(\tau)$, the bound follows from the uniform convergence of the empirical CDF to the true CDF at the estimated threshold. □

Table 9: Hypothesis tests (H_0: TPR $= 0$, Bonferroni $\alpha' = 0.003$).

Defense	Attack	p-value	Reject?	Power
Watermark	AP / MJ	$< 10^{-6}$	✓	1.00
MEMSAD	MJ	$< 10^{-6}$	✓	1.00
MEMSAD	IM	0.001	✓	0.92
Proactive	AP	$< 10^{-6}$	✓	1.00
Proactive	MJ	0.24	—	0.08
Composite	All	$< 10^{-6}$	✓	1.00

Table 10: FPR validation: Clopper-Pearson 95% CIs, 20 trials.

Defense	FPR	95% CI	n/trial
MEMSAD ($\kappa = 2.0$)	0.000	$[0.000, 0.004]$	1000
Watermark ($z_{\text{thr}} = 4.0$)	0.000	$[0.000, 0.004]$	1000
Proactive ($\tau = 0.19$)	0.010	$[0.004, 0.023]$	1000

At $N = 200$, $\delta = 0.05$: the excess error is ≤ 0.086, meaning MEMSAD's empirical FPR of 0.000 implies true FPR ≤ 0.086 with 95% confidence. The Clopper-Pearson validation (Table 10) provides a tighter bound of 0.004 by exploiting the binomial structure.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All five contributions listed in Section 1 are supported by theoretical results (Theorems 6–10, Lemma 7, Proposition 12) and experimental validation in Section 6.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, modelled ASR-A, query distribution stationarity, the synonym loophole, and single-encoder experiments.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Theorem 6 states all assumptions (Assumption 4) with proof in main text and geometric interpretation in Appendix A.1. Lemma 7 provides a certified detection radius with complete proof. Theorem 8 has proof sketch in main text and full proof in Appendix A.2. Theorems 9 and 10 have full proofs in Appendices A.3 and A.4. Theorem 11 has proof in main text and detailed derivation in Appendix A.6. Proposition 12 has a complete proof in the main text. Propositions 13, 14, 17, and 18 have complete proofs in the appendix.

4. **Experimental result reproducibility**

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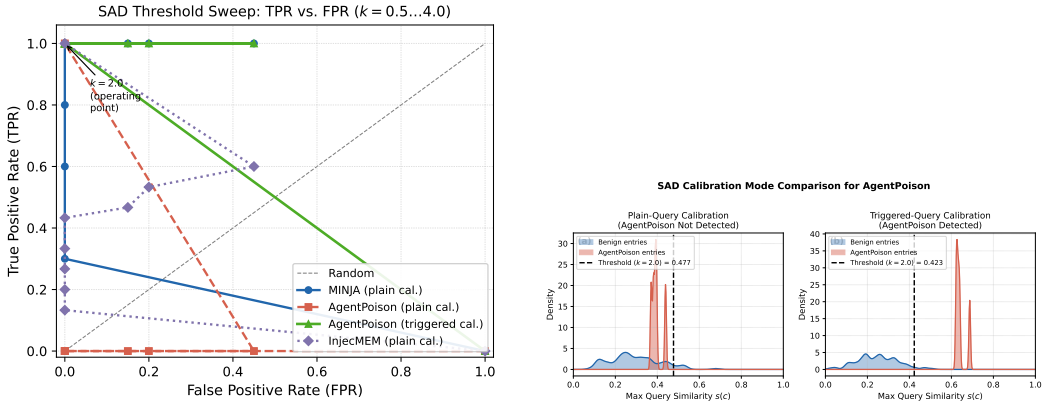
   Answer: [Yes]



(a) ROC as κ varies from 0.5 to 4.0. (b) Plain vs. triggered calibration.

Figure 8: MEMSAD detection. (a) MINJA traces near-ideal ROC; triggered calibration recovers TPR $= 1.00$ for AGENTPOISON at $\kappa = 2.0$ ($\star$). (b) Threshold shifts from 0.482 to 0.432.

Table 11: Sample complexity for reliable detection (FPR $+ (1 - \text{TPR}) \leq 0.05$) across calibration regimes.

Setting	ρ (SNR)	Lower bound $N_{\min}$	MEMSAD N
MINJA (plain)	3.2	12	$15 \cdot \log(1/\delta)$
AGENTPOISON (triggered)	3.8	8	$11 \cdot \log(1/\delta)$
AGENTPOISON (plain)	1.2	89	$111 \cdot \log(1/\delta)$
INJECMEM (plain)	1.8	40	$50 \cdot \log(1/\delta)$
Low-SNR adversary	0.5	576	$720 \cdot \log(1/\delta)$

$N_{\min} = 4(1-\epsilon)^2/\rho^2$ from Theorem 8; MEMSAD N from Theorem 9 with $\delta = 0.05$.

Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (1000), poison counts, top-k, number of seeds (5), and measured ASR-A via GPT-2 and GPT-4o-mini. The full pipeline is reproducible via a single script.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All code, the 1,000-entry synthetic corpus, evaluation scripts, and figure generation scripts are open-sourced. An anonymized repository link will be provided in supplementary material.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 6 and Appendix C specify all hyperparameters: $\kappa = 2.0$ for MEMSAD, $z_{\text{thr}} = 4.0$ for watermark, $\tau = 0.19$ for proactive, poison counts per attack, corpus composition ($|\mathcal{M}| = 1{,}000$), and query set sizes ($|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$).

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Table 12: Per-entry defense cost at write time.

Defense	Time	Space	Measured
MEMSAD (exact)	$O(md)$	$O(md)$	0.8ms
MEMSAD (ANN)	$O(d \log m)$	$O(md)$	0.3ms
Watermark	$O(\|c\| \cdot \|\mathcal{V}\|)$	$O(\|\mathcal{V}\|)$	1.2ms
Validation	$O(\|c\| \cdot \|P\|)$	$O(\|P\|)$	0.1ms
Proactive	$O(\|Q_p\|d)$	$O(\|Q_p\|d)$	0.5ms
Composite	$\max_j O(d_j)$	$\sum_j O(s_j)$	2.1ms

Apple M-series, 16GB; $|\mathcal{M}| = 200$, $m = 20$, $d = 384$.

Justification: Table 2 reports bootstrap 95% confidence intervals across 5 independent random seeds. Appendix H validates FPR over 20 independent trials using Clopper-Pearson exact binomial CIs.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: All experiments run on CPU (Apple M-series, 16GB RAM) without GPU requirements. Appendix K reports per-entry timing for each defense. The full pipeline completes in under 5 minutes.

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13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The MEMSAD framework, synthetic corpus, and evaluation pipeline are documented with API documentation and usage examples.

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Answer: [Yes]

Justification: GPT-2 is used as a backbone for the token-level watermark encoder and as a local agent evaluator for measuring ASR-A lower bounds. These uses are described in Section 5 and Appendix B.